Homework 5

ORF 245 - Fundamentals of Statistics

Due: Mar. 28th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Law of Large Numbers)
Let $X_1, ..., X_n \overset{iid}{\sim} \text{Poisson}(2)$.

(a) Define new random variables $Y_1, ..., Y_n$, such that

$$Y_i = \begin{cases} 1 & \text{if } X_i = 2 \\ 0 & \text{otherwise.} \end{cases}$$

What is the joint distribution of $Y_1, ..., Y_n$?

(Note: Another way we could define Y_i is: $Y_i = 1\{X_i = 2\}$. The function "$1\{\cdot\}$" equals one if the condition inside the brackets is true, and zero otherwise. You will encounter this notation later in this problem.)

- **Solution**: Since $X_i \sim \text{Poisson}(2)$, we have

$$\mathbb{P}(Y_i = 1) = \mathbb{P}(X_i = 2) = \frac{2^2 e^{-2}}{2!} = 2e^{-2}, \quad \mathbb{P}(Y_i = 0) = 1 - 2e^{-2}.$$

So each $Y_i \sim \text{Bernoulli}(2e^{-2})$. Since the X_i are i.i.d., and the transformation $X_i \mapsto Y_i$ is applied independently, the Y_i are also i.i.d. Bernoulli$(2e^{-2})$.
Thus, the joint distribution of $Y_1, ..., Y_n$ is the product of n independent Bernoulli$(2e^{-2})$ random variables.

(b) As $n \to \infty$, what value does $\bar{Y}_n := \frac{1}{n}\sum_{i=1}^n Y_i$ approach? Why?

- **Solution**: By the Law of Large Numbers, since the Y_i are i.i.d. with mean $\mathbb{E}[Y_i] = \mathbb{P}(X_i = 2) = 2e^{-2}$, we have

$$\bar{Y}_n := \frac{1}{n}\sum_{i=1}^n Y_i \xrightarrow{a.s.} 2e^{-2} \quad \text{as } n \to \infty.$$

(c) More generally, if $Z_i = 1\{X_i \in [a, b])\}$, for some two values of a and b, what value will $\bar{Z}_n$ approach as $n \to \infty$?

- **Solution**: Here Z_i is again an indicator random variable that equals 1 when $X_i \in [a, b]$, and 0 otherwise.
  Let $p = \mathbb{P}(a \le X_i \le b) = \sum_{k=\lceil a \rceil}^{\lfloor b \rfloor} \frac{2^k e^{-2}}{k!}$ (since $X_i \sim \text{Poisson}(2)$ is discrete).
  Then each $Z_i \sim \text{Bernoulli}(p)$, and by the Law of Large Numbers,

$$\bar{Z}_n := \frac{1}{n} \sum_{i=1}^{n} Z_i \xrightarrow{a.s.} p = \mathbb{P}(a \le X_i \le b) \quad \text{as } n \to \infty.$$

Exercise: 2. (Simulation: Law of Large Numbers)

Let $X_1, ..., X_n \overset{iid}{\sim} \text{Exp}(\frac{1}{2})$. The marginal pdf of each X_i is hence given by:

$$f_X(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}}, & 0 \le x < \infty \\ 0, & \text{else} \end{cases} \tag{1}$$

$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$ denotes the random sample mean as usual.

(a) For what values of n will we have $\text{Var}(\bar{X}_n) \le 0.01$?

- **Solution**: We are given that $X_i \sim \text{Exp}\left(\frac{1}{2}\right)$, so the mean and variance of each X_i are:

$$\mathbb{E}[X_i] = \frac{1}{\lambda} = 2, \quad \text{Var}(X_i) = \frac{1}{\lambda^2} = \frac{1}{(1/2)^2} = 4.$$

  Since $X_1, \ldots, X_n$ are i.i.d., the variance of the sample mean is:

$$\text{Var}(\bar{X}_n) = \frac{\text{Var}(X_i)}{n} = \frac{4}{n}.$$

  We want:

$$\frac{4}{n} \le 0.01 \quad \Rightarrow \quad n \ge \frac{4}{0.01} = 400.$$

  Answer: Any sample size $n \ge 400$ will ensure that $\text{Var}(\bar{X}_n) \le 0.01$.

(b) Perform the following simulation in R – for every n between $n = 1$ and $n = 2000$:

  (i) Generate realizations $x_1, ..., x_n$ of the joint random variables $X_1, ..., X_n$.
  (Recall these are assumed to be i.i.d. from an Exponential distribution with parameter $\frac{1}{2}$.)

  (ii) Compute the corresponding realization of the sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ and store it.

Plot the following two lines on the same figure. What do you observe and why?

- The realizations of the sample mean $\bar{x}_n$ vs the sample size n.
- The expected value of the random sample mean $\mathbb{E}[\bar{X}_n]$ vs n. *(Hint: This should be a horizontal line.)*

To do this exercise you may need to look up: 1) how to do for loops in R; 2) how to generate random samples from an exponential distribution.

- **Solution**: We simulated 2000 sample means from an Exp(1/2) distribution, where each sample mean $\bar{X}_n$ is based on a sample of size n. The resulting plot is shown in Figure 1.

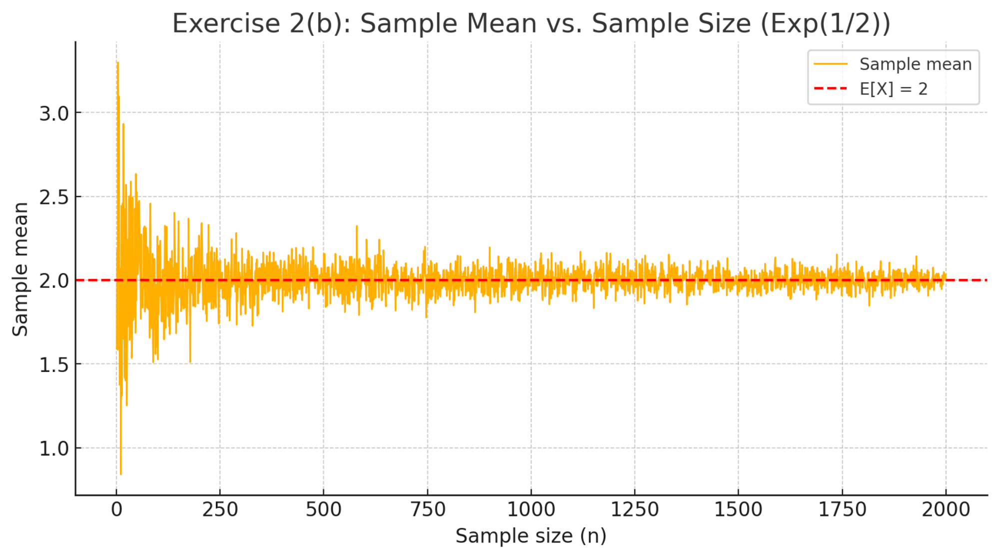

Figure 1: Realized sample means $\bar{X}_n$ versus sample size n, overlaid with the expected value $\mathbb{E}[X] = 2$.

The blue curve represents the realized sample mean $\bar{X}_n$ as n increases. The red dashed line corresponds to the theoretical expectation:

$$\mathbb{E}[X_i] = \frac{1}{\lambda} = 2.$$

We observe that the sample mean fluctuates significantly for small n, but becomes increasingly stable and close to 2 as n grows larger. This behavior is a demonstration of the **Law of Large Numbers**, which states that $\bar{X}_n \to \mathbb{E}[X]$ as $n \to \infty$.

Exercise: 3. (Simulation: Central Limit Theorem)
Let $X_1, ..., X_n \overset{iid}{\sim} \text{Exp}(\frac{1}{2})$, and $\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$ denotes the random sample mean as usual.

(a) How many samples n are needed to guarantee that the variance of $\bar{X}_n$ is no larger than 0.01?

- **Solution**: We know that for $X_i \sim \text{Exp}(1/2)$:

$$\text{Var}(X_i) = \frac{1}{(1/2)^2} = 4.$$

Then:

$$\text{Var}(\bar{X}_n) = \frac{4}{n}.$$

We want $\frac{4}{n} \leq 0.01 \Rightarrow n \geq \frac{4}{0.01} = 400.$
Therefore, we need at least $n = 400$.

(b) Let $n = 1000$. In R, generate realizations $x_1, ..., x_{1000}$ of the i.i.d. random variables $X_1, ..., X_{1000}$. Plot the histogram (of the values $x_1, ..., x_{1000}$), with bins of any size ensuring the resulting figure is informative.

What does this histogram look like? Why?

- **Solution**: We generated 1000 i.i.d. samples from the Exp(1/2) distribution. The histogram of these values is shown in Figure 2.

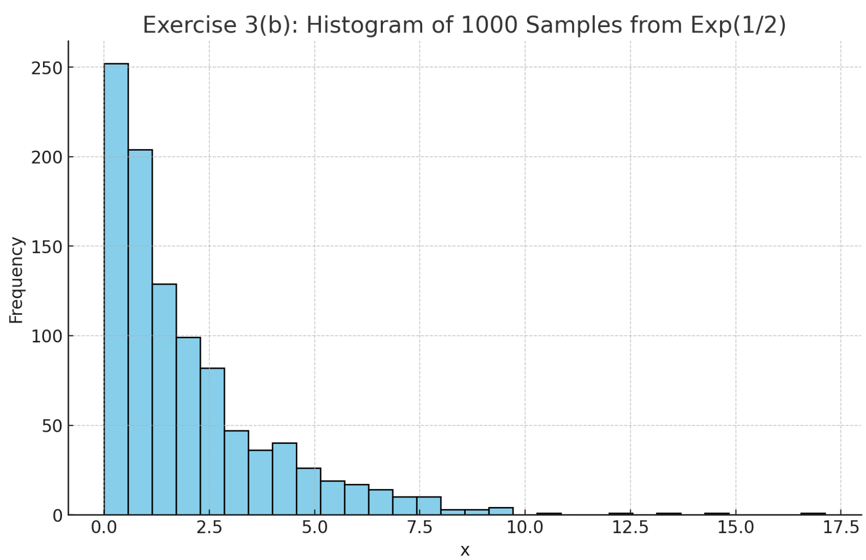

Figure 2: Histogram of 1000 samples from Exp(1/2) distribution.

The histogram is highly right-skewed, as expected for an exponential distribution. Most of the sample values are concentrated near 0, and the frequency decays rapidly as the values increase.

This shape reflects the probability density function of the exponential distribution:

$$f_X(x) = \frac{1}{2}e^{-x/2}, \quad x \geq 0.$$

The long right tail and high concentration of small values are consistent with the exponential distribution's theoretical behavior.

(c) Let $n = 1000$. We can obtain a single realization of the random sample mean $\bar{X}_{1000}$ by computing $\bar{x}_{1000} = \frac{1}{1000}\sum_{i=1}^{1000} x_i$, where $x_1, ..., x_{1000}$ are realizations of the joint random variables $X_1, ..., X_{1000}$.

Generate 5000 realizations of $\bar{X}_{1000}$ and plot the histogram of these 5000 values. Again, you may choose bins however you like as long as the resulting figure remains informative.

What does this histogram look like? Why? What is the approximate distribution of $\bar{X}_{1000}$?

- **Solution**: We generated 5000 independent realizations of the sample mean $\bar{X}_{1000}$, where each realization was computed as the mean of 1000 i.i.d. samples from an Exp(1/2) distribution.

  The resulting histogram of the 5000 sample means is shown in Figure 3.

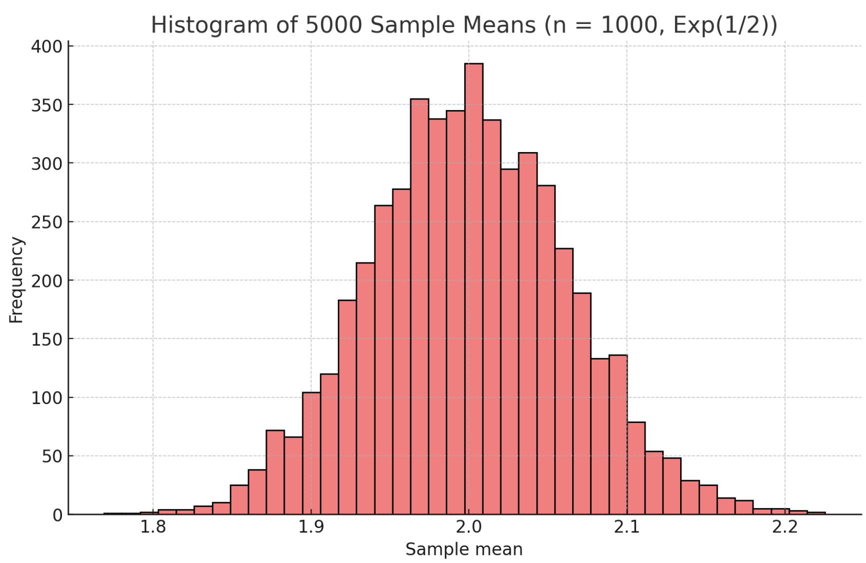

Figure 3: Histogram of 5000 realizations of $\bar{X}_{1000}$, each based on 1000 i.i.d. Exp(1/2) samples.

The histogram is bell-shaped and symmetric, centered around 2. This is expected because for $X_i \sim \text{Exp}(1/2)$, the mean is:

$$\mathbb{E}[X_i] = \frac{1}{\lambda} = 2.$$

Even though the exponential distribution is highly skewed, the sample mean of a large number of i.i.d. random variables is approximately normal, due to the Central Limit Theorem.

So:

$$\bar{X}_{1000} \approx \mathcal{N}\left(2,\ \frac{4}{1000}\right) = \mathcal{N}(2, 0.004),$$

where the variance comes from $\text{Var}(X_i) = \frac{1}{\lambda^2} = 4$.

Exercise: 4. (Central Limit Theorem)

Let $X_1, ..., X_{100} \overset{iid}{\sim} \text{Ber}(0.2)$, and define as usual $\bar{X}_{100} := \frac{1}{100}\sum_{i=1}^{100} X_i$.

(a) Compute $\mathbb{P}\left(\bar{X}_{100} > 0.22\right)$, in terms of the standard normal CDF $\Phi(\cdot)$.

- **Solution**: We use the Central Limit Theorem: since $X_i \sim \text{Bernoulli}(0.2)$, we have:

$$\mathbb{E}[X_i] = 0.2, \quad \text{Var}(X_i) = 0.2(1-0.2) = 0.16.$$

Then:

$$\bar{X}_{100} \approx \mathcal{N}\left(0.2, \frac{0.16}{100}\right) = \mathcal{N}(0.2, 0.0016).$$

We standardize:

$$\mathbb{P}(\bar{X}_{100} > 0.22) \approx \mathbb{P}\left(Z > \frac{0.22 - 0.2}{\sqrt{0.0016}}\right) = \mathbb{P}(Z > 0.02/0.04) = \mathbb{P}(Z > 0.5).$$

Therefore,

$$\mathbb{P}(\bar{X}_{100} > 0.22) \approx 1 - \Phi(0.5).$$

(b) Compute $\mathbb{P}\left(\left|\bar{X}_{100} - 0.2\right| > 0.02\right)$, in terms of the standard normal CDF $\Phi(\cdot)$.

- **Solution**: We again use the CLT:

$$\mathbb{P}\left(|\bar{X}_{100} - 0.2| > 0.02\right) = \mathbb{P}\left(|Z| > \frac{0.02}{\sqrt{0.0016}}\right) = \mathbb{P}(|Z| > 0.5).$$

Since this is a two-sided probability:

$$\mathbb{P}(|Z| > 0.5) = 2 \cdot \mathbb{P}(Z > 0.5) = 2(1 - \Phi(0.5)).$$

(c) Compute $\mathbb{P}\left(\bar{X}_{100} > 0.22\right)$ **without** using the standard normal CDF. How different is your answer from that of part (a)? Why?

(Hint: What is the distribution of $\sum_{i=1}^{100} X_i$, if $X_1, ..., X_{100}$ are iid Bernoulli random variables?)

- **Solution**: The sum $S := \sum_{i=1}^{100} X_i \sim \text{Binomial}(n = 100, p = 0.2)$. We have:

$$\mathbb{P}(\bar{X}_{100} > 0.22) = \mathbb{P}\left(S > 22\right) = \mathbb{P}(S \geq 23).$$

This can be computed using the cumulative distribution function of the binomial distribution, for example in R:

`1 - pbinom(22, size = 100, prob = 0.2)` This is equal to 0.26107.
Our answer in a) is $1 - \Phi(0.5) = 0.3$.
These two values are reasonably close. The difference is due to the fact that the CLT is only an approximation, and although $n = 100$ is moderately large, it's not large enough to make the normal approximation perfectly accurate, especially in the tail.